

torch.accelerator.memory.memory_stats

https://pytorch.org/docs/stable/generated/torch.accelerator.memory.memory_stats

Description:

Return a dictionary of accelerator device memory allocator statistics for a given device index. The return value of this function is a dictionary of statistics, each of which is a non-negative integer. "allocated.{all,large_pool,small_pool}.current,peak,allocated,freed": number of allocation requests received by the memory allocator. "allocated_bytes.{all,large_pool,small_pool}.current,peak,allocated,freed": amount of allocated memory.

Function Signature

```
torch.accelerator.memory.memory_stats(device_index=None, /)
[source]#
```

Parameters

Return type

Returns:

OrderedDict[str, Any]

Detailed Documentation

Documentation

Return a dictionary of accelerator device memory allocator statistics for a given device index.

The return value of this function is a dictionary of statistics, each of which is a non-negative integer.

"allocated.{all,large_pool,small_pool}.{current,peak,allocated,freed}": number of allocation requests received by the memory allocator.

"allocated_bytes.{all,large_pool,small_pool}.{current,peak,allocated,freed}": amount of allocated memory.

"segment.{all,large_pool,small_pool}.{current,peak,allocated,freed}": number of reserved segments from device memory allocation.

"reserved_bytes.{all,large_pool,small_pool}.{current,peak,allocated,freed}": amount of reserved memory.

"active.{all,large_pool,small_pool}.{current,peak,allocated,freed}": number of active memory blocks.

"active_bytes.{all,large_pool,small_pool}.{current,peak,allocated,freed}": amount of

active memory.

"inactive_split.{all,large_pool,small_pool}.{current,peak,allocated,freed}": number of inactive, non-releasable memory blocks.

"inactive_split_bytes.{all,large_pool,small_pool}.{current,peak,allocated,freed}": amount of inactive, non-releasable memory.

For these core statistics, values are broken down as follows.

all: combined statistics across all memory pools.

large_pool: statistics for the large allocation pool (as of June 2025, for size $\geq$ 1MB allocations).

small_pool: statistics for the small allocation pool (as of June 2025, for size $<$ 1MB allocations).

current: current value of this metric.

peak: maximum value of this metric.

allocated: historical total increase in this metric.

freed: historical total decrease in this metric.

In addition to the core statistics, we also provide some simple event counters:

"num_alloc_retries": number of failed device memory allocation calls that result in a cache flush and retry.

"num_ooms": number of out-of-memory errors thrown.

"num_sync_all_streams": number of synchronize_and_free_events calls.

"num_device_alloc": number of device memory allocation calls.

"num_device_free": number of device memory free calls.

device_index (torch.device, str, int, optional) – the index of the device to target. If not given, use torch.accelerator.current_device_index() by default. If a torch.device or str is provided, its type must match the current accelerator device type.

OrderedDict[str, Any]